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Handout for first-year medical students

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Shoulder Joint Complex

PLAN

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Goals

- Know how to classify a joint*
- Know the different articular surfaces that form a joint, and their means of union.*
- Know the different axes and movements of the joint.*

I -INTRODUCTION

The shoulder is a joint system that connects the upper limb to the trunk via the shoulder girdle. It includes a joint complex formed by five joints:

- 1- Scapulohumeral joint.
- 2- Acromioclavicular joint.
- 3- Sterno-costoclavicular joint.
- 4- Interscapulothoracic joint (false joint).
- 5- Subdeltoid joint (false joint)

Scapulohumeral Joint

II. DESCRIPTIVE ANATOMY

1- Definition:

It is a diarthrosis of the enarthrosis variety (spheroid). It joins the scapula to the humerus. It is a very mobile but fragile joint.

2/JOINT SURFACES:

- 1- The glenoid cavity of the scapula

Oval in shape. It is located at the level of the superolateral angle of the scapula. It has the glenoid tubercle in the center and the glenoid notch, on the anterior edge. It is surrounded on its peripheral edge by the glenoid labrum.

2- The humeral head

It occupies the proximal epiphysis of the humerus. It is rounded in shape, representing 1/3 of a sphere with 30 mm in diameter. Its surface is covered with a covering cartilage. Its axis forms an angle of 130° with that of the diaphysis.

3- The glenoid labrum

It is a fibrocartilaginous ring applied to the perimeter of the glenoid cavity and intended to increase the depth of this cavity. Seen in section, it is triangular, so we can recognize three faces: peripheral, external or articular, and internal adherent.

*The peripheral face :extends outward the surface of the neck of the scapula.

*The external or articular face: free, smooth and concave, is a part of the glenoid articular surface.

*The internal or adherent face: is closely connected to the periphery of the glenoid cavity.

3/ MEANS OF UNION:

A- The joint capsule

It is a thin fibrous sleeve of great laxity allowing a separation of the articular surfaces of 2 to 3 cm in extent. It is inserted on the one hand, on the peripheral face of the glenoid labrum , and on the other hand, on the anatomical neck of the humerus.

B- Passive ligaments

1- The coracohumeral ligament

This is the strongest ligament. It arises on the lateral edge of the coracoid process. It ends in two bundles: one on the greater tubercle (trochiter), and the other on the lesser tubercle (trochin) of the humerus.

2- The glenohumeral ligaments

There are three of them: the superior, middle and inferior glenohumeral ligaments.

***The superior glenohumeral ligament**

It arises from the glenoid labrum and the upper part of the neck of the scapula ,and ends on the upper part of the lesser tubercle (trochin) of the humerus.

***The middle glenohumeral ligament**

It originates in front of the previous one ,and ends on the lower part of the lesser tubercle (trochin) of the humerus.

***The inferior glenohumeral ligament**

It originates on the glenoid labrum and the neck of the scapula, and ends below the surgical neck of the humerus.

NB: These three ligaments delimit two spaces between them:

*Weitbrecht's foramen ovale: between the superior and middle glenohumeral ligaments.

*The foramen of Rouvière: between the middle and inferior glenohumeral ligaments.

3- The coracoglenoid ligament

This is the deep bundle of the coracohumeral ligament. It is located in the posterior region of the joint. It is stretched from the posterior part of the lateral edge of the coracoid process to the glenoid labrum.

3- Active ligaments

These are the tendons of the periarticular and mobilizing muscles of the scapulohumeral joint, these are the muscles: Supraspinatus , infraspinatus , subscapularis , teres minor, teres major, pectoralis major, latissimus dorsi, deltoid and coracobrachialis.

4- The synovium

It lines the deep surface of the joint capsule up to its insertions, where it is reflected to extend to the cartilaginous covering of the articular surfaces.

The synovial fluid produced by the synovial membrane ,provides lubrication and nutrition to the articular cartilage. The synovial membrane is lifted at the level of the humeral neck by the capsular frenulum (frenula capsulae).

5- The peri-articular bursae

There are a number of bursae between the capsule and the periarticular muscles whose role is to facilitate

sliding, the most important of which are:

- The subscapular bursa.
- The bicipital bursa.
- The subdeltoid bursa.
- The subcoracoid bursa.
- The bursa of the infraspinatus muscle.

III- FUNCTIONAL ANATOMY

The scapulohumeral joint is fragile but very mobile, possessing three degrees of freedom according to the 3 planes of movement and 3 axes:

In the frontal plane. Around the anteroposterior axis

*Adduction: It brings the limb inwards, pure adduction is not possible, associated with a retro-pulsion or ante-pulsion, can reach 40°.

*Abduction: It carries the limb outwards, the amplitude 90° without the participation of the scapulothoracic joint, it can reach 180° with the participation of the latter.

In the sagittal plane. Around the transverse axis:

*Flexion (ante-pulsion): It projects the limb forward, the amplitude 50-80° without the participation of the scapulothoracic joint, can reach 180° with the participation of the latter.

*Extension (retro-pulsion): It projects the limb backwards and never exceeds 30°, but can reach 40° following abduction and posterior sliding of the scapula.

In the transverse plane. Around a vertical axis

The forearm being flexed at 90° to eliminate pronation-supination:

***Lateral (external) rotation:** is limited to 30° amplitude.

***Medial (internal) rotation:** is limited to 80°, an anterior tilt of the scapula completes it to 90°.

***Circumduction:** It encompasses all types of movements and describes a cone with an apex passing through the center of the humeral head.

Sternocostoclavicular joint

- It is a joint by reciprocal interlocking: 1st rib, sternal end of the clavicle, and the sternal manubrium.

a. Articular surfaces

- Clavicular surface : covered with cartilage, it occupies the medial end of the clavicle
- Sternal surface: less extensive than the sternal surface.
- Costal surface: it is a narrow facet occupying the sternal end of the 1st rib, encrusted with cartilage

b. Means of union:

- **The capsule:** It is inserted around the perimeter of the 3 articular surfaces present.
- **The ligaments:** 4 in number reinforce the capsule:
 - Anterior and posterior ligaments
 - Superior ligament
 - Inferior ligament (or costoclavicular ligament)

c- Sliding means: There are 2 synovial membranes.

d. Joint physiology:

2 types of movements are possible: elevation-lowering, anterior and posterior projection;

Acromioclavicular joint

It is a flat-surfaced joint, or arthrodia, which joins the acromial end of the clavicle to the acromion

a. **Articular surfaces:**

Acromial surface: Elliptical, located on the medial edge of the acromion

- Clavicular surface: On the lateral end of the clavicle

b. **Means of union:**

- The capsule: It fits around the edge of the joint.
- Ligaments: They are divided into 2 groups:
 - the acromioclavicular ligament
 - the coracoclavicular ligaments. There are four of them:
 - Ligt trapezoid
 - Ligt conoid
 - **Coracoclavicular ligaments (medial and lateral).**

c. **Sliding means:** The synovium lines the deep surface of the capsule.d. **Joint physiology:**

The acromioclavicular joint is mobile in all 3 planes of space.

It has low mobility and transmits the movements of the sternoclavicular joint to the scapula.

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